1University of Westminster

School of Electronics and Computer Science

6MMCS009W Usability Testing and Evaluation Specification (2025/26)

Module leader	Dr Daphne Economou
Unit	Group Coursework [CW1]
Weighting	Group Coursework [CW1] 50%
Qualifying mark	30%
Description	**Group coursework** [CW1] - assesses the students' skills in: forming a research question, organising, planning and conducting user testing and collecting data that help addressing the research question.
Learning Outcomes Covered in this Assignment	**Group Coursework [CW1]** LO1 Demonstrate an applied understanding of the role of evaluation in usability context LO2 Demonstrate knowledge and understanding in usability testing and evaluations by selecting appropriate methods and techniques in a variety of contexts LO3 Plan, organise and conduct appropriate testing and evaluation methods for selected products and systems
Handed Out	6th October 2025
Due Date	**Group Coursework [CW1] – 20th November 2025, 1pm Online**
Expected deliverables	**Group Coursework [CW1]** • A report that consolidates all the required documents that need to be produced to prepare and conduct the usability test • Spreadsheet with the collected data • Evidence of the data collection process • Part-A-Research-Ethics-Consideration-Application-form2020_21
Method of Submission	Electronic submission on Blackboard
Type of Feedback and Due Date	**Formative Feedback:** • Present progress on the Group Coursework [CW1] to your tutor/s by weeks 7 & 8 during tutorial/seminar time to get instant formative feedback and update your report before the final submission Group Coursework [CW1] **will be presented at a viva during tutorial time in week 10** and students will receive instant feedback. **Summative feedback (marks and written feedback)** on both CW will be provided within 3 weeks after the submission (mark on Blackboard, additional oral feedback in class and discussion of model answers). All marks will remain provisional until formally agreed by an Assessment Board.

Assessment regulations

Refer to Part 3 Assessment regulations for taught courses guide for undergraduate students for a clarification of how you are assessed, penalties and late submissions, what constitutes plagiarism etc.

https://www.westminster.ac.uk/current-students/guides-and-policies/academic-matters/academic-regulations
Handbook of Academic Regulations 2024/25 (PDF)

Penalty for Late Submission

If you submit your coursework late but within 24 hours or one working day of the specified deadline, 10 marks will be deducted from the final mark, as a penalty for late submission, except for work which obtains a mark in the range 40 – 49%, in which case the mark will be capped at the pass mark (40%). If you submit your coursework more than 24 hours or more than one working day after the specified deadline you will be given a mark of zero for the work in question unless a claim of Mitigating Circumstances has been submitted and accepted as valid.

It is recognised that on occasion, illness or a personal crisis can mean that you fail to submit a piece of work on time. In such cases you must inform the Campus Office in writing on a mitigating circumstances form, giving the reason for your late or non-submission. You must provide relevant documentary evidence with the form. This information will be reported to the relevant Assessment Board that will decide whether the mark of zero shall stand. For more detailed information regarding University Assessment Regulations, please refer to the following website: http://www.westminster.ac.uk/study/current-students/resources/academic-regulations

Group Coursework description [CW1]

The purpose of this group coursework (CW1) is to design and conduct a usability testing study for a chosen product (software/service). The design of the experimental setup in usability testing involves several key components to ensure that the study is systematic, valid, and capable of addressing the research questions.

1. Research Questions and Hypotheses

Research Questions: Each group member should create a unique research question guiding the study (1 research question) **focusing on a different usability metrics** for the same application/product/service.

Hypotheses: Each group member should formulate one hypothesis that corresponds to the research question they created, detailing expected relationships between variables. **Each hypothesis should align with one of the following statistical tests**:

- **Independent-sample t-test**
- **Paired-sample t-test**
- **ANOVA**
- **Pearson correlation**

As a result, the team should collectively have **four research questions**, **each with a hypothesis that corresponds to one of the statistical tests listed above**.

Each group member should also **describe each hypothesis correctly mathematically**.

2. Variables and Measures

Each group member clearly describes the variables for each formed hypothesis:

- **Independent Variables**
- **Dependent Variables**
- **Control Variables**
- **Measurement Scales**: Define scales for collecting data (e.g., Likert scales, time, other).

3. Experimental Design

a. Experimental design type

As a group, describe the **test method used** (e.g., between-subjects, within-subjects, or mixed design/split-plot). Each team member should then explain how this experimental design supports the data collection needed to evaluate their individual hypothesis.

b. Procedure

As a group determine **where** the test will be conducted (e.g., in person, remotely, or in a naturalistic setting) and if will be **moderated or un-moderated** with justification about your choice.

As a group collaboratively create **a test session flow/plan**, outlining step-by-step procedures, including instructions, task order, data collection methods, and timing. This test session flow should facilitate the collection of data for all the hypotheses. Each team member should contribute specific steps related to the hypothesis they formulated.

c. Tasks and Scenarios

Each team member creates and contributes to the group a **task** the study participants should follow that represents typical or critical actions within the product (e.g. navigating a menu or completing a purchase) that allows the collection of data to evaluate their individual hypothesis.

Create realistic scenarios for each task to provide context and encourage natural user interaction.

d. Roles & responsibilities

As a group outline **the roles** of those involved in the usability test (e.g., moderators, facilitators) and explain how each group member contributed to meeting the study's requirements and facilitating the sessions effectively.

e. Planning

As a group explain what **planning** you had to do to prepare for the Usability Testing session (e.g. lab bookings, arrange for laptops or set up devices with the necessary software).

f. Data collection methods

As a group identify **tools you will use for accurate data collection and session recording** (e.g. screen recording software, teams for remote moderated test setting) and justify why those are best fit for the test session you are designing.

Each group member should explain how you collect the data that corresponds to the hypothesis you have formulated.

g. Research Instruments/Test material

As a group:

- Prepare a **prototype** to be studied (it could be an interactive prototype using a rapid prototyping tool), or **identify and describe** the **software**, **website or service you are testing**.
- **Prepare** a questionnaire or specify other methods (e.g., software logs) you will use to collect the required data. Each team member must explain and provide evidence of the materials they use to collect data relevant to evaluating their individual hypothesis (e.g. which question of the user questionnaire collects data for their individual hypothesis).
- **Participant Information Sheet** clearly outline the purpose of the study, procedures, duration, confidentiality, and voluntary nature (check the examples provided on Blackboard).
- **Participant Consent Form** asking the participants to agree to the study and data storage, ensuring compliance with data protection standards (check the examples provided on Blackboard).
- Create consistent **instructions for participants**, ensuring that they understand each task without influencing their actions.
- Create consistent **instructions for moderators**, if needed, explaining their role in the test session.

4. Participants

Define **characteristics of participants** relevant to the study, like age, expertise level, or familiarity with similar products.

Describe the participant **recruitment process** and provide evidence (e.g. email that was sent to recruit participants).

Collect data from at least 20 participants. If using remote methods, ensure instructions are clear and accessible and the method you are using to record the data allows collecting data with precision.

5. Ethical Considerations

As a group, fill out and **submit the research ethics** form to ensure your study aligns with ethical guidelines, covering participant privacy, data handling, and consent.

Part-A-Research-Ethics-Consideration-Application-form2020_21

6. Study results (Data)

Provide evidence for the collection of data from 20 participants (e.g. the participants' responses, or a fraction of some test session recordings complying to General Data Protection Regulation (GDPR)). This should be o**riginal data files, such as exported CSV files from software logs**, with participant information anonymized but still traceable to the session.

Present the collected data in a spreadsheet, corresponding to the evaluation of each hypothesis formulated by the group.

Identify the statistical methods suitable for analyzing your data based on your hypotheses and data types (e.g., t-tests, ANOVA, correlation).

You should also collect and submit as a group **evidence for the data collection process**, this could be:

- **Screenshots or Photos of Data Collection in Progress** showing participants interacting with the interface or completing tasks, along with timestamps for authenticity.
- **Timestamped Software logs**, timestamped entries, or system-generated records that show real-time data points (e.g., clicks, errors, or completion times).
- **Video Recordings snippets of usability sessions**, highlighting task completion steps, participant feedback, or interactions.
- **Scans or digital copies of completed questionnaires**, with participant responses to both quantitative and qualitative questions.
- **Observer Notes** taken during sessions, including observed user behavior, comments, and notable events.
- For remote tests, **screenshots of data from analytics or remote usability testing platforms** can help verify participant interaction.
- If tests were conducted in person, **a participant sign-in sheet with timestamps** can serve as proof of session attendance.

7. Report Writing and Data Presentation

As a group, prepare a report detailing the entire process of creating and conducting the usability test. Each member should contribute content for the tasks they managed. Include a critical reflection on the overall process, discussing successes, challenges, and areas for improvement. Additionally, submit a spreadsheet with the collected data, clearly labelled with variables and results for each hypothesis.

Follow the provided template.

The report should be approximately 3000 words excluding the appendix, although this is not a strict word count.

8. Prepare for Oral Examination (Viva)

As a group you present the entire process you followed to create the usability testing plan and to conduct the study in an oral examination (viva).

Presentation structure: Walk through the process step-by-step, highlighting each member's role and contribution.

Key Points: Emphasize research questions, hypotheses, experimental design, data collection, analysis, and insights.

What and how to submit Group Coursework submission [CW1]

Submit as a group:

1. **a common report** that:
   - describes the design process of the usability testing study for a chosen product (software/service) with clear justification of your decisions
   - consolidates all the required documents (participant information sheet, participant consent form, instructions for participants, instructions for moderators, questionnaires or other way of collecting data required to be produced to prepare and conduct the usability test (use the provided report template)
2. **Spreadsheet with the raw data**
3. **Evidence of the data collection process addressing all the hypotheses**
4. **Part-A-Research-Ethics-Consideration-Application-form2020_21** (use the provided form)

The expected report contribution: Each group member is responsible for contributing content to the report, covering all tasks for which they were individually responsible.

- **Group Responsibilities**: Together, the group is responsible for producing a comprehensive study plan, defining study tasks, creating a participant information sheet, designing the questionnaire, recruiting

participants, and collecting data for all the research questions developed. Additionally, the group should document the entire preparation process, describe how the usability test was conducted, and summarize the outcomes. A critical reflection on the overall process should also be included.

- **Individual Responsibilities**: Each group member should provide a clear outline of their tasks related to the preparation and conduct of the usability test.

Group Coursework [CW1] marking scheme

This assignment is marked based on the following marking criteria.

	Group coursework [CW1] requirements	I/G/GI	Mark
1	**1 Research question per student** focusing on evaluating usability metrics (clear, focused (1.5 marks), and testable (1.5 marks)research question that aligns with usability testing objectives (2marks))	I	5
1	**1 hypothesis for each research question that evaluates 1 usability metric and serves:** • **independent-sample-test** • **paired-sample t-test** • **ANOVA** • **Pearson correlation** (clear, focused (1.5 marks), and testable (1.5 marks) hypothesis that aligns with usability testing objectives (2marks))	I	5
1	**Describe each hypothesis correctly mathematically**, use appropriate symbols and notation (3 marks) to express the hypothesis clearly	I	3
2	**Clearly describe the variables for each formed hypothesis** (dependent (2 marks), independent (2 marks), control variables (2 marks) and measure scale (2 marks))	I	8
3a	**Experimental design type** (between-subjects, within-subjects, or mixed design/split-plot) clearly identify the appropriate experimental design type for each hypothesis formed (2 marks) and justification (2 marks), smooth integration in the group task (3 marks)	GI	7
3b	**Procedure** (in person, remote, moderated, or un-moderated) clear description (3 marks) and justification (3 marks)	G	6
3c	**Tasks and Scenarios** (task the study participants should follow that allows the collection of data to evaluate individual's test hypothesis)	I	3
3d	**Roles & Responsibilitie**s (outline the roles of those involved in the usability test (e.g., moderators, facilitators) (1 marks) and provide Instructors Information Sheet (3 marks)	G	4
3e	**Planning for the Usability Session** explain the planning to prepare for the Usability Testing session (2 marks)	G	2
3f	**Data collection methods** (identify tools you will use for accurate data collection and session recording (3 marks) and justify why those are best fit for the test session you are designing (3 marks))	G	6
3g	**Research Instruments/Test material** (prototype, questionnaires or description of other ways data is collected (3 marks), instructions for moderators (2 marks)), how this is measured (e.g. if you record task time define task start and end) (2 marks)	GI	7
4	**Participants** (define participants characteristics (2 marks), describe the participant recruitment process (2 marks), provide evidence for recruiting 20 participants (3 marks))	GI	7
5	**Ethical Considerations - Complete and submit Part-A**-Research-Ethics-Consideration-Application-form2020_21 (one as a group) (4 marks) **Participant information sheet** explaining the test process to the participants and a consent form asking the participants' consent to use and store the collected data (3 marks)	G	7
6	**Study Results (Data)** (raw data from 20 participants in a table (3 marks) organised per hypothesis tested (2 marks), evidence for the data collection process (5 marks)	GI	10

7	**Report Writing and Data Presentation** (Clarity and Organization (1 mark), Adherence to Guidelines (1 mark), Completeness and Detail (1 mark), Use of Evidence and Data Presentation (1 mark), Critical Analysis and Reflection (1 mark), Writing Quality and Style (1 mark)) **Remove instructions from the template (4 marks)**	**GI**	10
8	**Prepare for Oral Examination (Viva)** (Clarity and Structure (2 marks), Content Coverage (2 marks), Data Presentation and Visual Aids (2 marks), Delivery, Engagement and Time Management (2 marks), Q&A Preparedness, Professionalism (2 marks))	**GI**	10
	Group coursework [CW1] Total		**100**

I	Individual marking
G	Group marking
GI	Evidence of individual's contribution to the group task